

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 326 — Past Questions & Materials

by Renance

Course: PEE 326

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 300

This study pack was compiled and restructured by Renance for PEE 326 students at Federal University of Petroleum Resources, Effurun. It brings together 4 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 4 — PEE 326 Oil and Gas Pollution and Control

Level 300 • First Semester • PastQuestion

- FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

2023/2024 Session: Second Semester Examination

PEE326: OIL AND GAS POLLUTION AND CONTROL

Answer Question 1 and any other 3 Time: 2 Hours

Please write your Matriculation Number Boldly on Your Script

Question one

1a) Bos, a science magazine of the American Geophysical Union, notes that 90% of the world's oil spills are caused by humans, not nature, What are the most common accidental, human-originating oil spill causes?

1ai) what do you understand by "bilge water discharge"?

1aii) what are the two notable international oil spills and their causes?

1b) what are the causes of hulls damage?

1c) what is the duty of the government when a global temperature rise and concerns around climate

change continue to become more dire.

v Question two

2a) what is a kick?

2ai) when a kick is not properly handled what happens

2aii) what are the causes of a kick and (2ai)

2b) mention and explain the factors responsible for pipeline leakage

Question three .

3ai) Enumerate three measures used to tackle pipeline and flowline leakage in oil and gas high pressured line

3aii) what are the likely hindrance pipeline leakage can cause in the oil and gas industry such as water, nitrogen, and hydrocarbons, which can further

3b) Sour gases contain impurities, gases and their effect?

complicate processing and transportation, what are those

3c) Devise a safety protocol to tackle sour gas production as a petroleum Engineer.

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Question four

4a) mention three potential environmental concerns of sour gas spills

4b) what is the national and international organization responsible for the regulatory framework for transporting sulfur gas

4c) what are the sustainable alternative shift towards a cleaner future

V Question five

5a) List and explain the mechanical methods of removing oil and gas pollution

5ai) write short note on the advantages and disadvantages of mechanical removal method of oil spill. :

-) mention two chemical methods of oil and gas pollution removal

5bi) At least mention one advantage and disadvantage of chemical method of oil and gas pollution.

6) Enumerate biological methods of removing oil and gas pollutant

6i) List the advantage and disadvantage of biological method of oil and gas pollutant, \ £.

Paper 2 of 4 — PEE 326 OIL AND GAS POLLUTION AND CONTROL 2024I2025

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
 2023/2024 Session: Second Semester Examination
PEE326: OIL AND GAS POLLUTION AND CONTROL

Answer Question 1 and any other 3

Time: 2 Hours

Please write your Matriculation Number Boldly on Your Script

Question one

1a) Eos, a science magazine of the American Geophysical Union, notes that 90% of the world's oil spills are caused by humans, not nature. What are the most common accidental, human-originating oil spill causes?

1ai) what do you understand by "bilge water discharge"?

1aii) what are the two notable international oil spills and their causes?

1b) what are the causes of hulls damage?

1c) what is the duty of the government when a global temperature rise and concerns around climate change continue to become more dire.

✓ **Question two**

2a) what is a kick?

2ai) when a kick is not properly handled what happens

2aii) what are the causes of a kick and (2ai)

2b) mention and explain the factors responsible for pipeline leakage

2c) List 3 types of pipeline and flowline leakage

Question three

3ai) Enumerate three measure used to tackle pipeline and flowline leakage in oil and gas high pressured line

3aii) what are the likely hindrance pipeline leakage can cause in the oil and gas industry

3b) Sour gases contain impurities, such as water, nitrogen, and hydrocarbons, which can further complicate processing and transportation, what are those gases and their effect?

3c) Devise a safety protocol to tackle sour gas production as a petroleum Engineer.

e no complete

*Continuous
intermittent (I don't get the spelling)*

catastrophic

You may suppose (Copy my Jotter later ☺)

Question four

- 4a) mention three potential environmental concerns of sour gas spills
- 4b) what is the national and international organization responsible for the regulatory framework for transporting sour gas
- 4c) what are the sustainable alternative shift towards a cleaner future

✓ Question five

- 5a) List and explain the mechanical methods of removing oil and gas pollution
- 5ai) write short note on the advantage and disadvantages of mechanical removal method of oil spill.
- 5aj) mention two chemical methods of oil and gas pollution removal
- 5bi) At least mention one advantage and disadvantage of chemical method of oil and gas pollution.
- 5cj) Enumerate 3 biological methods of removing oil and gas pollutant
- 5ci) What is the advantage and disadvantage of biological method of oil and gas pollutant.

Paper 3 of 4 — PEE 326 Oil & Gas Pollution Studies & Control

Level 300 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF PETROLEUM ENGINEERING

SEMESTER TEST

31* August, 2023.

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NY 4 QUESTIONS DURATION: 30 MINUTES

14(a). Mention any 7 pollution encountered in the oil and gas industry?

(b). Illustrate with diagram what you understand by water cycle?

(c). what are the main aqueous effluents in exploratory operations?

ial i ces?

(d). what are the potential impact to fresh-water source:

i i seismic survey?

2(a). Highlight five practice in planning a seism!

ction installations?

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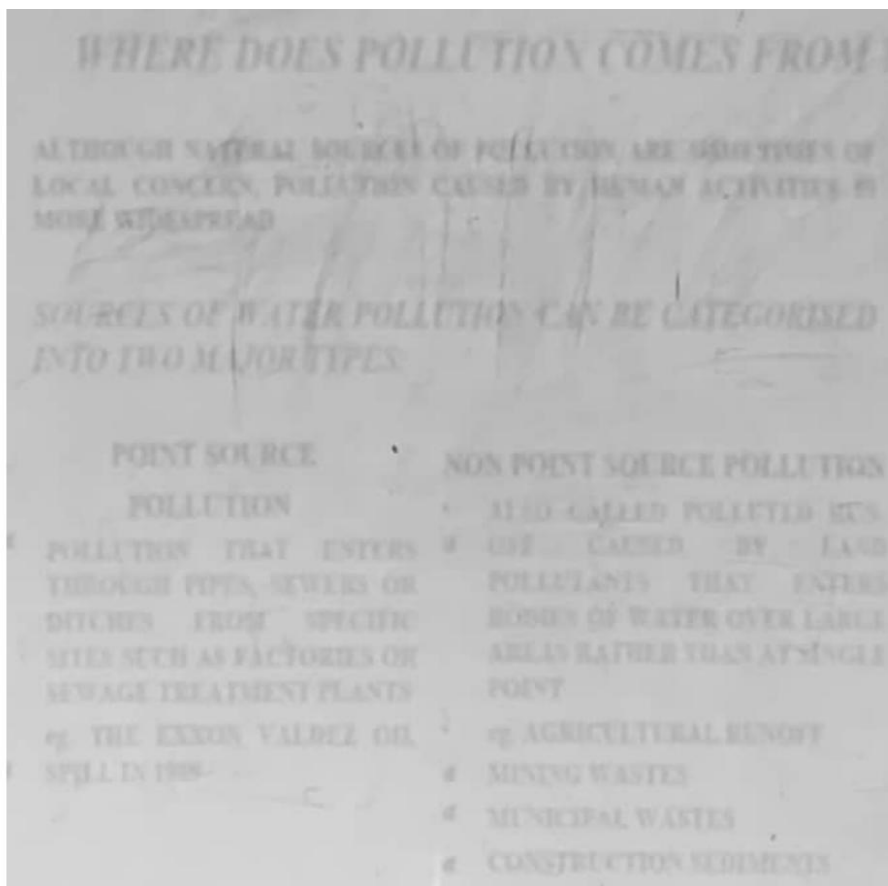
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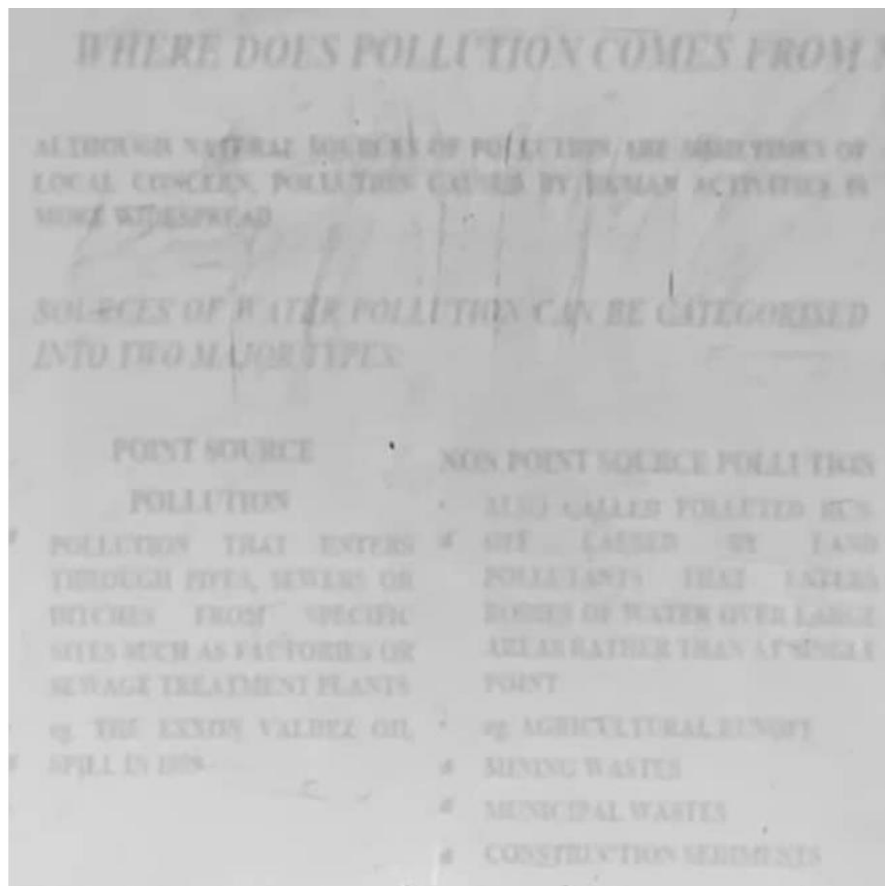
(b). what are tl major noise sources at prod!

Paper 4 of 4 — PEE 326

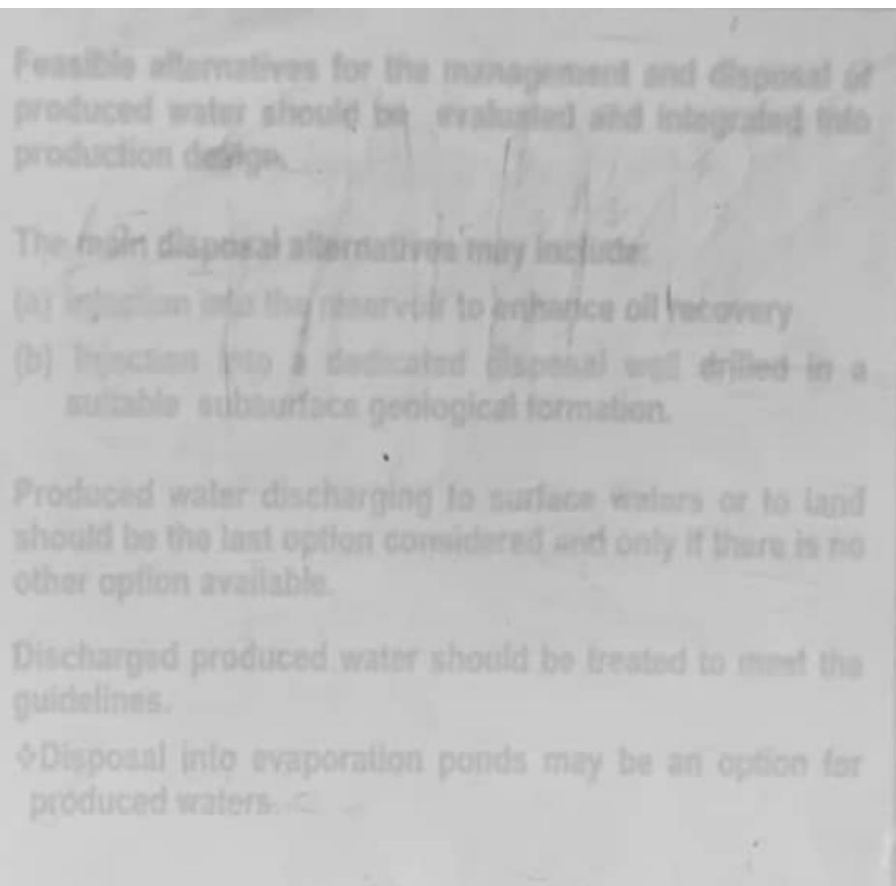
Level 300 • Second Semester • Handout

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Standards for Industrial Effluents			
	Maximum Level Surface Water	Maximum Level Ground Water	Area
Lead (mg/l)	0.01	-	1
Cadmium (mg/l)	1	-	2
Chromium (mg/l)	2	-	2
Copper (mg/l)	3	-	1
Zinc (mg/l)	5	-	15
Selenium (mg/l)	0.05	-	0.05
Nickel (mg/l)	1	-	5
Boron (mg/l)	2	-	-



Volumes of waste produced depend on the stage of the exploration and production process.

During seismic operations, waste volumes are minimal and relate mainly to vessel activities.

In exploratory drilling the main aqueous effluents are drilling fluids and cuttings.

In production operations – after the development wells are completed – the primary effluent is produced water.

Standards for Industrial Effluents		
Parameter	Into In Land Surface Water	Into Marine Coastal Area
Percent sodium		60
Cyanide mg/l	0.2	0.2
Chloride mg/l	1000	—
Fluoride mg/l	2	15
Dissolved phosphate mg/l	5	—
Sulphate mg/l	1000	—
Sulfide mg/l	2	5

Standards for Industrial Effluents		
Parameter	Discharge into Land Surface Water	Discharge into Marine Coastal Area
Suspended Solids(mg/l)	100	(a) For process waste water - 100 (b) For Cooling water effluents 10% above T-suspended matter of influents cooling water.
Particle size of suspended solids	Shall pass 850 micron IS Sieve	(a) Floatable solids max.3mm (b) Settleable solids max.850

Standards for Industrial Effluents		
Parameter	Into In Land Surface Water	Into Marine Coastal Area
Dissolved solids (Inorganic) mg/l max.	2100	-
pH value	5.5 to 9	5.5 to 0 45 at the point of discharge
Temperature °C	Shall not exceed 40 in any section of the stream within 15 mts. down stream from the effluents outlet	
Oil & Grease mg/l max.	10	20

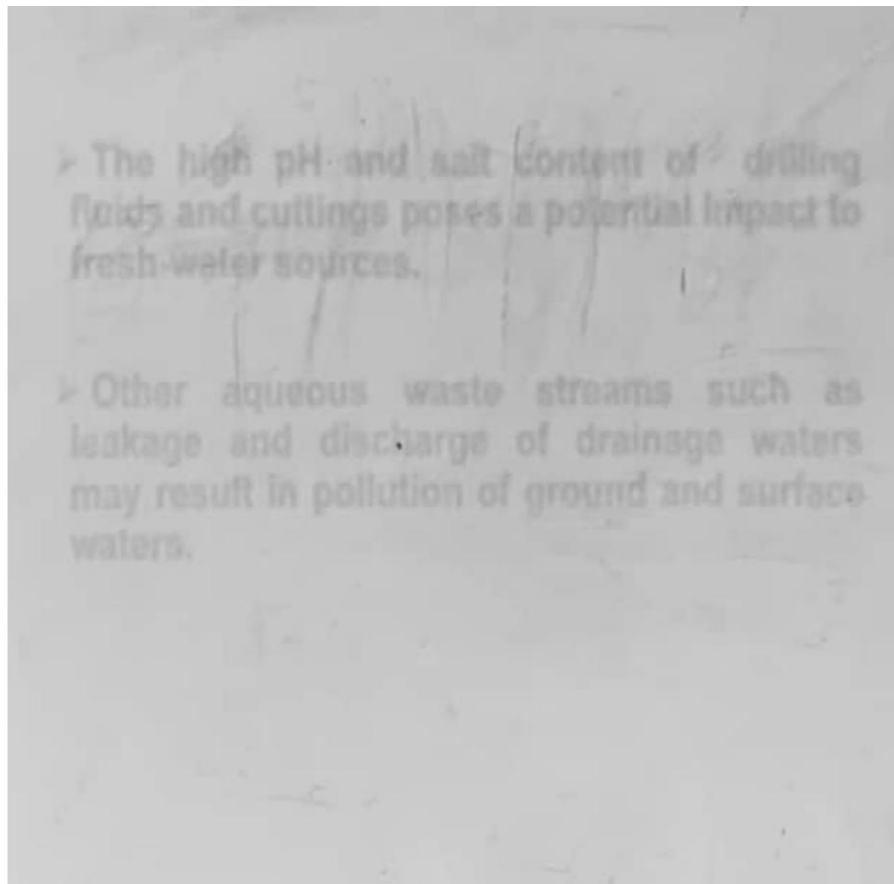
WATER POLLUTION

The principal sources of water pollution resulting from exploration and production operations are:

1. Produced Water.
2. Drilling fluids, Cuttings and well treatment chemicals.
3. Process, wash and Drainage water.
4. Sewerage, sanitary and domestic wastes.
5. Spills and leakages and
6. Cooling water.

PRODUCED WATER

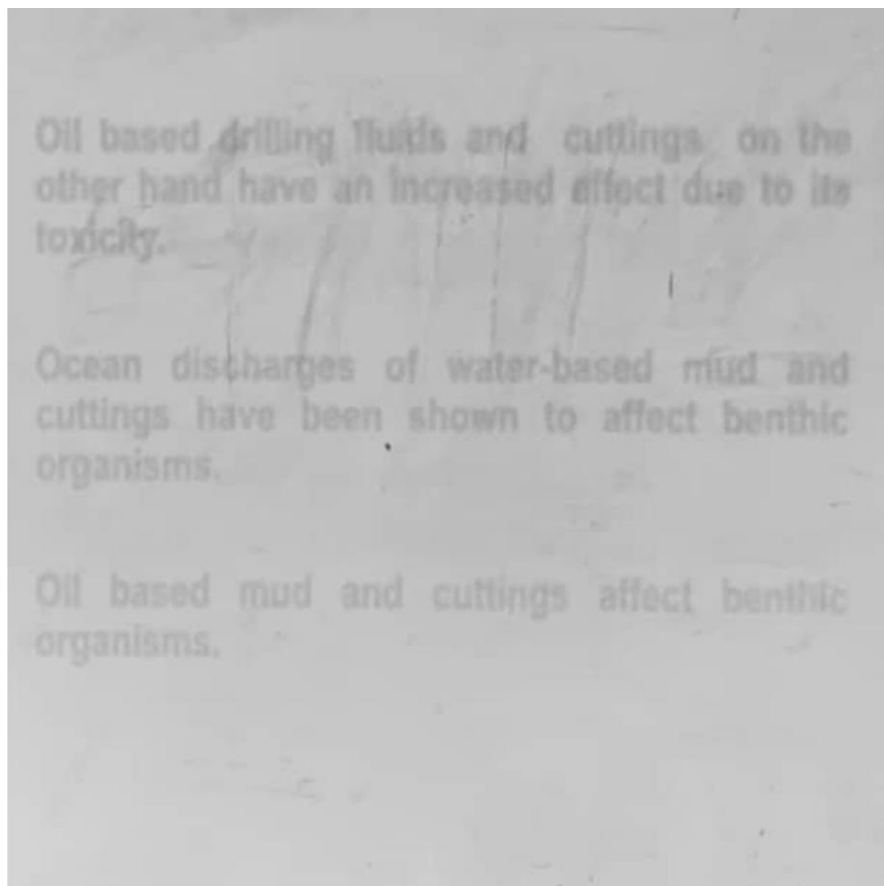
- Water coming out from reservoir, is separated from the oil and gas in the production facility.
- The produced water can be one of the largest waste products, by volume, which needs to be managed and disposed off by the oil and gas industry.
- Produced water contains a complex mixture of inorganic (dissolved salts, trace metals, suspended particles) and organic (dispersed and dissolved hydrocarbons, organic acids) compounds.



WATER POLLUTION

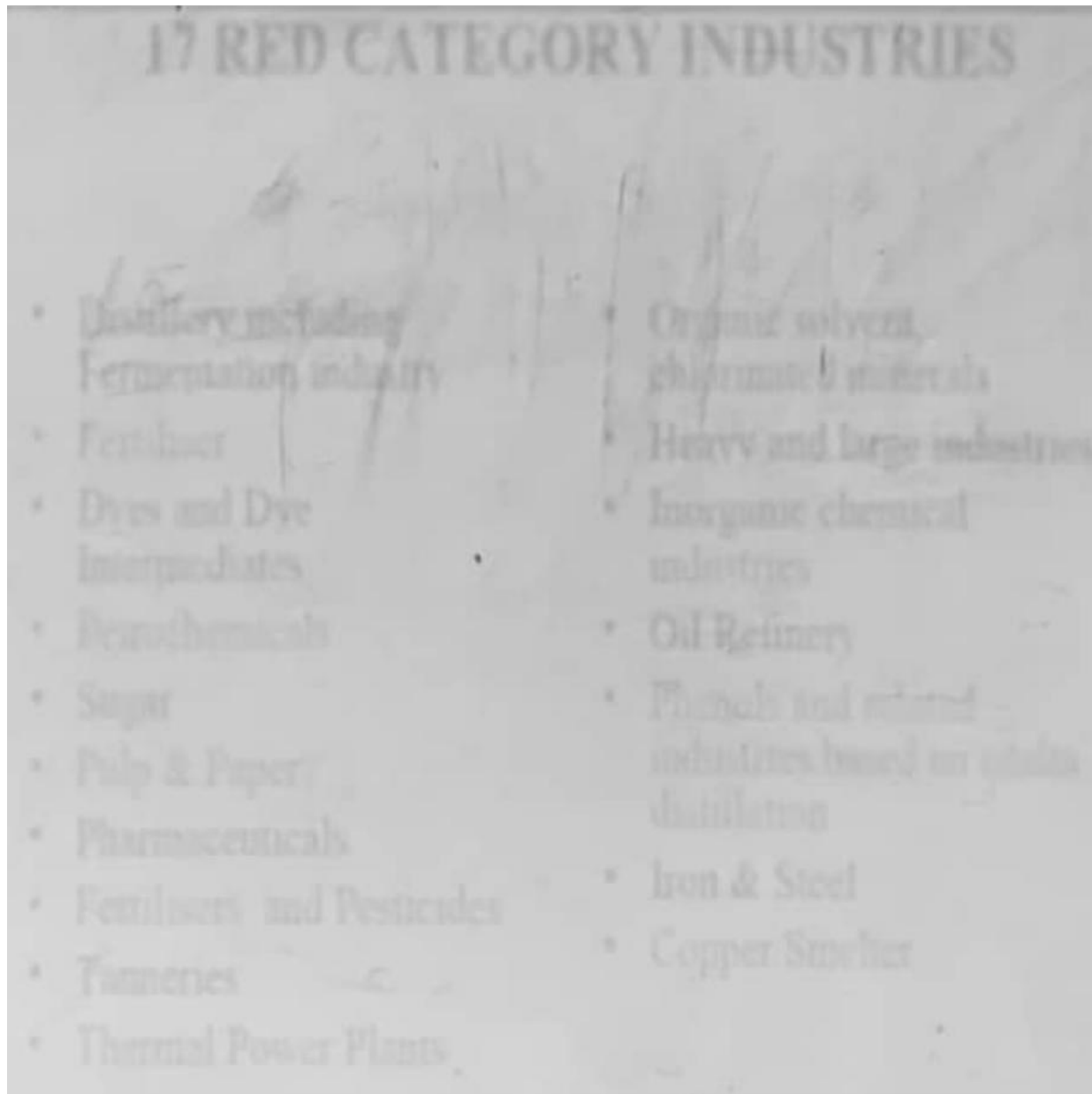
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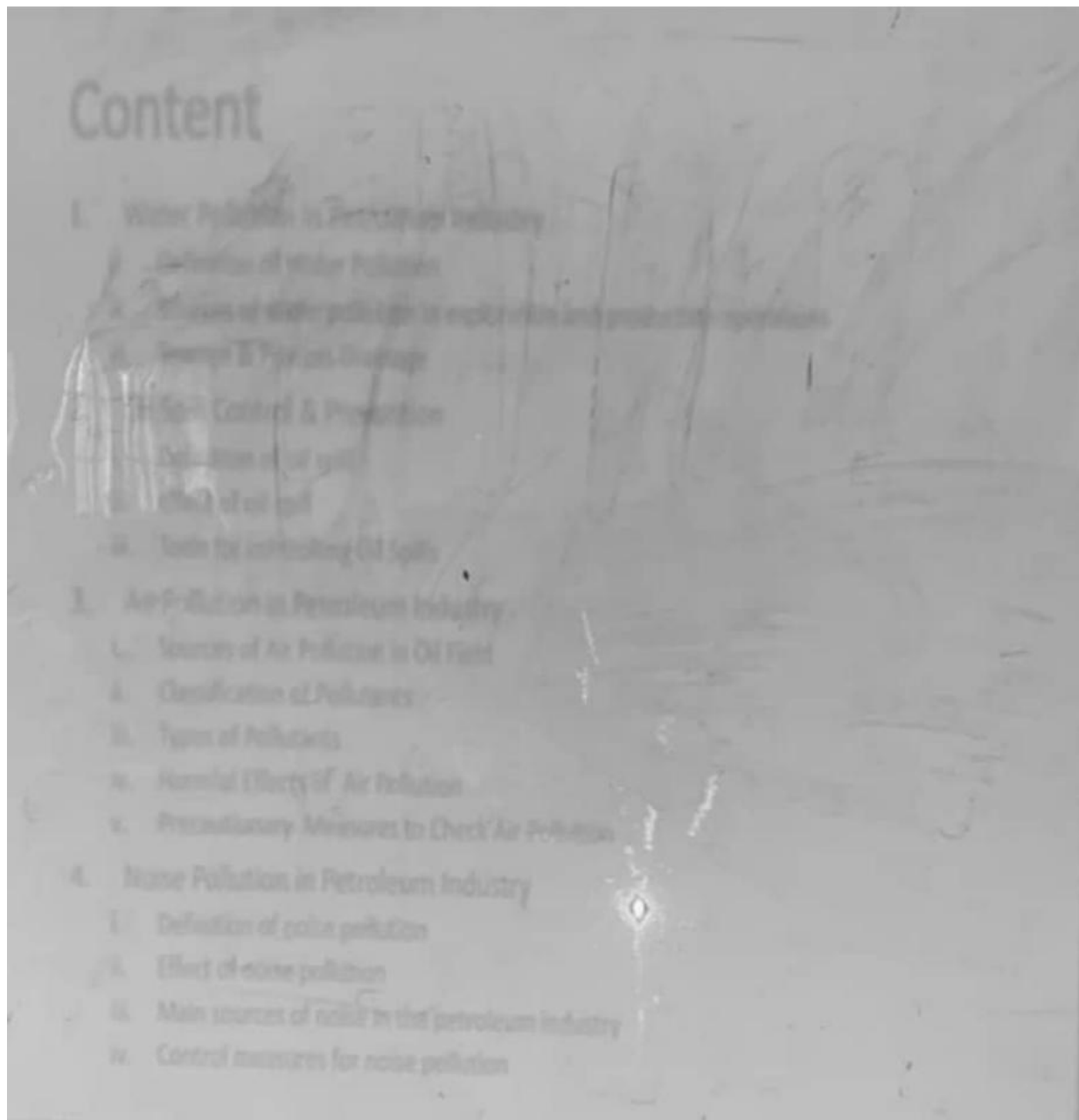


WATER POLLUTION

- Water pollution can be defined as presence of solid, liquid or gaseous contaminants in such concentration that may alter the quality of water.
- Water pollution is defined as the addition to water of an excess material or heat that is harmful to the living organism or which impairs the beneficial use of water.



Emissions, Effluent & Waste Levels from Onshore oil & Gas Development	
Parameter	Guideline Value
Produced Water	For discharge to surface waters to be left
	✦ Total HC content: 10mg/l
	✦ pH: 6 – 9
	✦ BOD: 25mg/l
	✦ COD: 125mg/l
	✦ TSS: 35mg/l
	✦ Phenols: 0.5mg/l
	✦ Sulfides: 1mg/l
	✦ Heavy metals: 5mg/l
	✦ Chlorides: 600mg/l (average) & 1200mg/l (maximum)



Standards for Industrial Effluents		
Parameter	Into Land Surface Water	Into Marine Coastal Area
Ammoniacal nitrogen (AsN) mg/l max.	50	50
Free ammonia	5	5
Biochemical oxygen demand	30	100
Chemical oxygen demand mg/l	250	250
Arsenic mg/l	2	2